

Computer science

HL and SL

Paper 1

26.9.2025

~~80~~ minutes

60

Instructions to candidates

- Do not open this examination paper until instructed to do so.
- Answer all questions.
- The maximum mark for this examination paper is **[30 marks]**.

1. Write down **one** keyword (reserved word) of Python. [1]
2. Identify **two** advantages of storing your program code using a version control system (such as git) together with an online storage service (such as GitHub). [2]
3. Outline the operation of an interpreter. [2]
4. Consider the example below.

```
class Teacher:
    def __init__(self, name, subject):
        self.name = name
        self.subject = subject

wiltso = Teacher('Ville', 'mathematics')
print(wiltso.name + ', ' + wiltso.subject)
```

When the program is run, the output is:

Ville, mathematics

Explain the difference between a class and an object, using the example above to illustrate your explanation. [3]

5. Consider a function with name `intolerable` ("sietämätön") whose parameter is an integer temperature `t`, and which will return `True` if the value of the parameter is at most `-40` or at least `50`, and will return `False` otherwise.
 - (a) Write down the signature of the function. [2]
 - (b) Write the function. [3]

6. Write a program which will output the smallest positive integer divisible by both 387 and 6381. [4]

7. Consider the program below. The program should output True if at least one element of nums is negative. Otherwise the output should be False.

```
# output True if there is at least one negative element in list nums,  
# otherwise output False  
nums = [4, 3, -1, 5, -2, 1]
```

```
found = False  
for n in nums:  
    if n < 0:  
        found = True  
    else:  
        found = False
```

```
print(found)
```

In this case the output of the program is

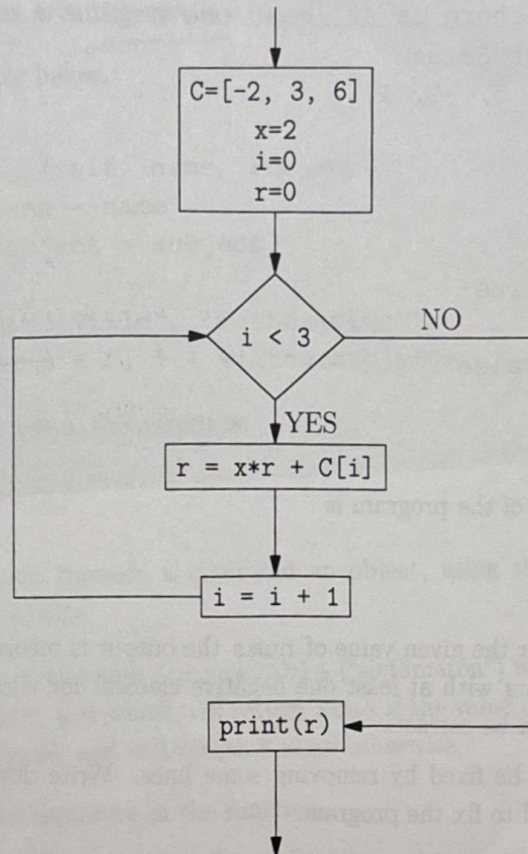
False

- (a) As can be seen, for the given value of nums the output is incorrect. Write down a value of list nums with at least one negative element for which the output of the program would be correct. [1]
- (b) The program can be fixed by removing some lines. Write down the lines that should be removed to fix the program. [2]
- (c) After fixing the program by removing some lines, it can still be made more efficient. Without writing any code, suggest how the program could be made more efficient. [2]

8. Copy and complete the trace table below for the given flowchart.

[4]

x	i	r	i < 3	C[i]	x*r+C[i]	output
:	:	:	:	:	:	:



9. Write a function

even_odd_indices(numbers)

The function returns a list where the elements of numbers with even indices, in their original order, are followed by the elements of numbers with odd indices, also in their original order. The contents of numbers is not modified. The table below shows examples of return values for different arguments (values of the parameter).

argument	return value
[0, 1, 2, 3, 4]	[0, 2, 4, 1, 3]
[-5, 4, 7, 2, 11, -9]	[-5, 7, 11, 4, 2, -9]
[1, 2]	[1, 2]
[99]	[99]

[4]

1.
 - if
 - while
 - etc.
2.
 - retaining code history
 - backup
 - sharing code
 - collaborating
3.
 - (a) read statements
 - (b) evaluate
 - (c) print result
4.
 - class defines a datatype
 - object is an instance of that type
 - in the example, Teacher is a datatype for teachers, while wiltsu is an object which corresponds to a specific teacher

5. (a) intolerable(t)

(b)

```
def intolerable(t):  
    if t <= -40 or t >= 50:  
        return True  
    else:  
        return False
```

~~print(intolerable(-50))~~

- condition
- return True
- return False

6. (various optimizations are available, here is a brute-force solution)

```
i = 1
found = False
while not found:
    if i % 387 == 0 and i % 6381 == 0:
        found = True
    else:
        i += 1
print(i)
```

274383

- variable and its incrementation
- loop with correct condition (or e.g. True and break)
- conditional with correct condition
- output

7. (a) any array whose last element is negative
(b) remove `else` and following line
(c) program can be made more efficient by stopping array traversal as soon as a negative value has been found

```
odds = [v for (i, v) in enumerate(numbers)
        if i % 2 == 1]
```

```
return odds + odds
```

```
numbers = [4, 3, -1, 5, -2, 1]
print(odd, odd, odds(numbers))
```

```
(4, -1, -2, 3, 5, 1)
```

1. correct even index copies

2. appended for similar

3. correct odd index copies

4. return value

8.

x	i	r	i<3	C[i]	x*r+C[i]	output
2	0	0	true	-2	-2	
2	1	-2	true	3	-1	
2	2	-1	true	6	4	
2	3	4	false			4

- values of i
- values of i<3
- values of x*r+C[i]
- output

9. • solution using append

```
def even_odd_indices(numbers):  
    result = []  
    i = 0  
    while i < len(numbers):  
        result.append(numbers[i])  
        i += 2  
    i = 1  
    while i < len(numbers):  
        result.append(numbers[i])  
        i += 2  
  
    return result
```

for i in range(0, len(numbers), 2)
→ result.append(numbers[i])

X { numbers = [4, 3, -1, 5, -2, 1]
print(even_odd_indices(numbers))

/input()

[4, -1, -2, 3, 5, 1]

- list comprehensions

```
def even_odd_indices(numbers):  
    evens = [v for (i, v) in enumerate(numbers)  
              if i % 2 == 0]  
    odds = [v for (i, v) in enumerate(numbers)  
            if i % 2 == 1]  
  
    return evens + odds
```

X { numbers = [4, 3, -1, 5, -2, 1]
print(even_odd_indices(numbers))

[4, -1, -2, 3, 5, 1]

1. correct even index copies
2. appended (or similar)
3. correct odd index copies
4. return value